

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I Pratip Raj(192521409) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pratip raj

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Q1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications. Analyze Performance, Security, and Manageability, then Justify the Better Choice.

Answer

A **hypervisor** is software that creates and manages multiple Virtual Machines (VMs) on a physical server.

Comparison of Type-1 and Type-2 Hypervisors

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware	Runs on top of a host operating system
Performance	Very high	Lower due to host OS overhead
Security	More secure	Less secure because host OS can be attacked
Resource Usage	Efficient	Consumes more resources
Reliability	High	Moderate
Examples	VMware ESXi, Microsoft Hyper-V, Xen VMware Workstation, Oracle VirtualBox	

Performance Analysis

- Type-1 directly communicates with hardware, reducing latency.
- It provides faster CPU, memory, and disk performance.
- Suitable for high-performance cloud applications.

Security Analysis

- Smaller attack surface because there is no host operating system.
- Better VM isolation.
- Supports secure boot, encryption, and role-based access control.

Manageability

- Supports centralized VM management.
- Enables live migration, snapshots, and automated resource allocation.
- Easier monitoring in large cloud data centers.

Justification

For **mission-critical cloud applications**, **Type-1 Hypervisor** is the best choice because it offers:

- High performance
- Strong security
- Better scalability
- High availability
- Efficient resource utilization

Conclusion

Type-1 Hypervisors are preferred for enterprise cloud data centers due to their superior performance, security, and reliability.

Q2. A Cloud Provider Must Isolate Workloads Belonging to Finance, Healthcare, and Student Portals on the Same Hardware. Analyze How Virtualization Architecture Supports Isolation and Identify Two Major Attack Surfaces.

Answer

Virtualization-Based Isolation

A hypervisor creates separate virtual machines for each department:

- Finance VM
- Healthcare VM
- Student Portal VM

Each VM has:

- Separate operating system
- Independent memory
- Separate storage
- Independent virtual network

Failure in one VM does not affect the others.

Isolation Techniques

- Hypervisor-based isolation
- Virtual LAN (VLAN)
- Virtual Firewalls
- Storage isolation
- Resource quotas

Major Attack Surfaces

1. Hypervisor Attack

- If attackers compromise the hypervisor, all virtual machines may be affected.

Protection

- Regular security patches
- Secure boot

- Access control

2. VM Escape Attack

- Malware inside one VM attempts to access another VM or the hypervisor.

Protection

- Strong VM isolation
- Updated hypervisor
- Continuous monitoring

Conclusion

Virtualization ensures secure workload isolation while protecting sensitive finance and healthcare applications from unauthorized access.

Q3. VM Host Utilization Snapshot

Given

- CPU Utilization = **92%**
- Memory Utilization = **88%**
- Storage I/O Latency = **35 ms**
- Average VM Boot Time = **170 seconds**

Answer

Analysis of Bottlenecks

1. CPU Bottleneck

CPU utilization of **92%** indicates heavy processing load.

Corrective Action

- Add more CPU cores.
- Enable auto-scaling.
- Migrate some VMs to another host.

2. Memory Bottleneck

Memory usage of **88%** is very high.

Corrective Action

- Increase RAM.
- Remove unused VMs.
- Optimize memory allocation.

3. Storage Bottleneck

Storage latency of **35 ms** is higher than the recommended value (<10 ms).

Corrective Action

- Upgrade HDD to SSD.
- Use faster storage arrays.
- Enable storage caching.

4. Slow VM Boot Time

170 seconds indicates poor storage and high resource contention.

Corrective Action

- Improve storage performance.
- Reduce CPU and memory contention.
- Optimize VM startup sequence.

Conclusion

The major bottlenecks are **CPU overload, high memory usage, slow storage, and long VM startup time**. Increasing resources and optimizing virtualization settings will improve performance.

Q4. Analyze How Software Defined Datacenter (SDDC) Improves Agility Compared with Traditional Datacenter.

Answer

A **Software Defined Datacenter (SDDC)** virtualizes compute, storage, and networking, allowing all infrastructure to be managed through software.

Compute Virtualization

- Multiple virtual machines share one physical server.
- Fast VM creation and migration.
- Better CPU utilization.

Storage Virtualization

- Combines multiple storage devices into one logical storage pool.
- Easy expansion.
- Automatic backup and replication.

Network Virtualization

- Creates virtual switches, routers, and firewalls.
- Faster network configuration.
- Improved security and flexibility.

Advantages over Traditional Datacenter

Traditional Datacenter	Software Defined Datacenter
Manual configuration	Automated management
Slow deployment	Rapid deployment
Fixed hardware	Virtual resources
High maintenance	Lower operational cost
Limited scalability	Highly scalable

Conclusion

SDDC provides **automation, flexibility, scalability, efficient resource utilization, and faster service deployment**, making it more agile than traditional data centers.

Q5. A Multi-Cloud Federation Project Fails Due to Identity Mismatches Between Providers. Analyze the Issue and Propose a Secure Identity and Privacy Design Using Federation Concepts.

Answer

Problem Analysis

Different cloud providers use different identity management systems.

This causes:

- Login failures
- User identity mismatches
- Authorization errors
- Security risks

Federated Identity Solution

Use a centralized Identity Provider (IdP) supporting:

- Single Sign-On (SSO)

- SAML
- OAuth 2.0
- OpenID Connect (OIDC)

Users authenticate once and securely access multiple cloud providers.

Privacy and Security Design

- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- End-to-end encryption
- Audit logs
- Least-privilege access
- Token-based authentication

Benefits

- One login for multiple clouds.
- Strong security.
- Easy user management.
- Better compliance.
- Improved privacy protection.

Conclusion

Federated identity management with **SSO, SAML/OAuth, MFA, RBAC, and encryption** provides secure authentication, protects user privacy, and ensures seamless access across multiple cloud providers.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: